# Chapter 1: Regions and Availability Zones

A region is a geographic area that contains one or more datacentres.
An availability zone is an isolated location within a region.
Zones have independent power, cooling and networking.

# Chapter 2: Virtual Machines

A virtual machine (VM) is an emulation of a physical computer.
VM scale sets let you run many identical virtual machines.
Resizing a virtual machine changes its CPU and memory.

# 2.1 Storage for Virtual Machines

Managed disks hold the data of a virtual machine.
Premium SSD disks offer the lowest latency for production workloads.

# Chapter 3: Networking

A virtual network (VNet) isolates your resources in the cloud.
A load balancer distributes traffic across healthy instances.
Network security groups filter traffic by port and address.

# Chapter 3 (continued)

Peering connects two virtual networks privately.
A VPN gateway links a network to an on-premises site.

# Summary

Regions contain zones. Zones are isolated. VMs run in zones.